8715 course project report: Kriging-informed Optimal Transport Conditional Flow Matching for Sea Level Anomaly Downscaling

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1 Introduction

Climate downscaling aims to reconstruct high-resolution regional climate fields from coarse-resolution observations or simulations. It plays an important role in Earth science, including ocean circulation analysis, extreme weather prediction, climate risk assessment, etc. However, climate variables often exhibit complex multi-scale spatial structures and strong non-stationary dynamics, making downscaling a challenging problem. In particular, a single coarse-resolution observation may correspond to multiple plausible fine-scale realizations, rendering climate downscaling an inherently ill-posed inverse problem.

Traditional statistical downscaling approaches, such as kriging and interpolation-based methods, rely on spatial smoothness assumptions to estimate fine-scale fields. While these methods are computationally efficient, they often over-smooth spatial structures and fail to capture complex nonlinear climate dynamics. More recently, deep learning approaches have demonstrated improved capability in modeling nonlinear spatial patterns. Nevertheless, they remain deterministic and struggle to adequately represent uncertainty and physical consistency in climate systems. Furthermore, diffusion-based generative models for climate downscaling typically require computationally expensive iterative sampling procedures, limiting their practical efficiency [1].

Recent advances in generative modeling, like flow matching, provide a promising alternative for probabilistic climate downscaling. Compared with diffusion models, conditional flow matching (CFM) learns continuous deterministic transport trajectories between source and target distributions, enabling faster sampling through ordinary differential equation (ODE) integration [2]. In particular, Optimal Transport Conditional Flow Matching (OT-CFM) [3] introduces optimal transport coupling to construct straighter and more stable probability paths, improving vector field learning efficiency and training stability.

Motivated by these advantages, we investigate the application of OT-CFM to sea level anomaly (SLA) downscaling and propose a kriging-informed optimal transport conditional flow matching framework, termed Ki-OT-CFM. The proposed method combines Universal Kriging interpolation with OT-CFM to incorporate spatial priors into the generative process. Specifically, kriging-interpolated conditional maps guide the vector field learning process, while variogram regularization is introduced to preserve spatial dependence structures characteristic of oceanographic fields. Compared with diffusion-based counterparts, the proposed framework enables faster training and inference while maintaining competitive reconstruction quality.

We evaluate Ki-OT-CFM on regional SLA downscaling over the East North America (ENA) region using satellite-derived sea level anomaly observations. Experimental results demonstrate that the proposed framework significantly improves over traditional interpolation-based approaches and achieves faster training and inference than diffusion-based methods while preserving fine-scale spatial structures.

The main contributions of this project can be summarized as follows:

- We investigate the application of optimal transport conditional flow matching to climate downscaling and propose the Ki-OT-CFM framework for sea level anomaly downscaling.
- We incorporate Universal Kriging interpolation and variogram regularization into the conditional flow matching framework to improve spatial coherence and preserve physically meaningful spatial structures [1].
- We demonstrate that Ki-OT-CFM achieves competitive reconstruction quality while reducing training and inference cost compared with diffusion-based generative approaches.

1.1 Problem Formulation

We study the task of downscaling for sea level anomaly (SLA) fields. Given a coarse-resolution observation $\tilde{y} \in \mathbb{R}^{H_c \times W_c}$, the objective is to generate a high-resolution field $x_0 \in \mathbb{R}^{H \times W}$ that is numerically consistent with observations while preserving realistic spatial structures.

Following prior work, we first apply Universal Kriging interpolation to obtain a dense conditional field

$$y = \mathcal{K}(\tilde{y}),$$

where $\mathcal{K}(\cdot)$ denotes the kriging interpolation operator. The resulting field y serves as a spatially coherent prior that guides the generative process.

Our objective is to learn the conditional distribution

$$p(x_0 \mid y)$$

using a conditional flow matching framework OT-CFM[2].

2 Related Work

2.1 Source Distribution Design in Flow-based Super Resolution(SR)

Some conditional flow-based SR methods adopt the conventional noise-to-data transport paradigm, where generation starts from Gaussian noise while low-resolution observations are incorporated as conditional guidance. Flow matching for Sentinel-2 super-resolution [4] formulates remote sensing SR as a conditional flow generation problem by concatenating LR Sentinel-2 inputs with noisy samples while initializing the transport process from Gaussian noise. Latent-FlowSR [5] further generates HR latent representations from Gaussian latent noise under LR latent conditions in a compressed latent space. SfmSR [6] additionally incorporates semantic guidance extracted from LR images into the flow matching process to improve reconstruction consistency while still relying on Gaussian-noise initialization. However, since these methods follow a noise-to-data transport formulation, where the source distribution is largely independent of degraded LR observations, they potentially lead to suboptimal transport trajectories and reduced structural consistency during reconstruction.

Recent flow-based SR methods argue that LR latent representations already lie close to the target HR manifold, and therefore directly model the transport process from degraded observations to HR distributions. FlowSR [7] reformulates SR as a rectified flow bridging LR and HR images, enabling efficient LR-to-HR trajectory learning for single-step restoration. FLowHigh [8] explores data-dependent probability paths by aligning the source distribution with LR audio distributions instead of using a standard Gaussian prior, improving convergence and reconstruction quality for audio super-resolution. Although these methods better exploit degradation-aware structures, directly transporting LR distributions toward HR targets may result in complex transport dynamics and suboptimal trajectory alignment. Allo$\{\text{SR}\}^2$ [9] further argues that directly replacing Gaussian noise with LR latents may introduce domain gaps and trajectory deviation, and therefore proposes SNR-guided trajectory initialization to align LR latent features with intermediate states on the pretrained generative flow. Nevertheless, the method depends on heuristic trajectory initialization and strong alignment to pretrained generative flows, potentially limiting flexibility and generalization under complex degradations.

2.2 Perceptual–Fidelity Trade-offs in Generative SR

Recent generative super-resolution methods can synthesize visually realistic details, but often suffer from hallucinated structures and degraded numerical fi-

delity, motivating the need to balance perceptual enhancement with structural, physical, or semantic consistency.

RareFlow [10] and SRast [11] both emphasized the importance of physical consistency in generative super-resolution beyond perceptual quality alone. RareFlow introduced physics-aware constraints and gated spatial conditioning to reduce physically inconsistent hallucinations in remote sensing super-resolution, while SRast employed physically constrained flow matching on the simplex to preserve local mass conservation and biological consistency in spatial transcriptomics super-resolution. Physics-constrained methods often rely on domain-specific priors and may overly restrict generative flexibility and perceptual detail synthesis.

Some recent generative Real-ISR methods employ reward-driven or preference-guided optimization to balance perceptual realism and fidelity preservation. OARS [12] proposed a quality-adaptive reward framework to dynamically balance perceptual enhancement and structural consistency according to input degradation severity, while LucidNFT [13] introduced a multi-reward RL framework jointly optimizing perceptual quality and LR-conditioned faithfulness to mitigate semantic and structural hallucinations caused by overly aggressive perceptual optimization. Reward-driven methods heavily depend on reward design and may still produce hallucinated details under imperfect fidelity supervision.

2.3 Generalization and Robustness in SR

Recent works improve SR generalization by reducing reliance on dataset-specific degradation statistics and learning transferable transport priors. SRast [11] decouples semantic representations from spatial geometry and aligns latent distributions across biological domains to achieve strong zero-shot generalization. RareFlow [10] further enhances OOD robustness under cross-sensor distribution shifts through physics-aware conditional generation and uncertainty-guided control. Nevertheless, these methods rely on domain-specific priors, target-domain adaptation, or handcrafted transport assumptions, which may limit generalization across unseen degradation distributions and domains.

2.4 Flow Matching in Downscaling

Flow matching has recently emerged as an effective framework for probabilistic climate downscaling and small-scale physics generation. WxFlow [14] applies conditional flow matching to transport Gaussian noise toward high-resolution precipitation distributions under coarse climate and topographic conditions. Subsequent works further improve distribution alignment and robustness by introducing latent-space conditioning and adaptive stochastic transport. SerpentFlow [15] separates large-scale shared climate structures from fine-scale variability through frequency-domain latent decomposition and generative do-

main alignment, while SFM [16] and AFM [17] encode coarse-resolution inputs into target-aligned latent distributions before applying adaptive flow matching to synthesize stochastic fine-scale physics under spatially misaligned PDE dynamics. More recently, PC-AFM [18] incorporates physics-aware conservation constraints into adaptive flow matching to improve physical consistency and geographic generalization under climate distribution shifts. Although these methods attempt to improve source-distribution alignment, physical consistency, and generalization, robust transport modeling under stochastic dynamics and distribution shifts remains challenging.

3 Methods

3.1 Overview

We propose Ki-OT-CFM, a kriging-informed conditional flow matching framework for stochastic sea level anomaly (SLA) downscaling. Figure 1 illustrates the overall architecture of the proposed method.

Algorithm 1 Training in Ki-OT-CFM

repeat
  $(x_0, \tilde{y}) \sim p(x_0, \tilde{y})$
  $y \leftarrow \text{U-Krig}(\tilde{y})$
  $t \sim \text{Uniform}([0,1])$
  $z \sim \mathcal{N}(0, I)$
  $(z, x_0) \leftarrow \pi_{\text{OT}}(z, x_0)$
  $x_t = (1-t)\, z + t\, x_0$
  $u_t = x_0 - z$
  $\hat{x}_0 = \dfrac{x_t - (1-t)\, z}{t}$
  Compute $\Delta_{\hat{x}_0}(\boldsymbol{h})$ and $\Delta_{x_0}(\boldsymbol{h})$
  $\mathcal{R}_V = \dfrac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \left(\Delta_{\hat{x}_0}(\boldsymbol{h}) - \Delta_{x_0}(\boldsymbol{h})\right)^2$
  Take a gradient descent step on

$$\nabla_\theta \Big(\|v_\theta(x_t, y, t) - u_t\|^2_{\mathbf{m}} + \lambda_V \mathcal{R}_V \Big)$$

  where $\|\cdot\|^2_{\mathbf{m}}$ is masked MSE over ocean mask $\mathbf{m}$
until converged

Algorithm 2 Inference in Ki-OT-CFM

$x_0 \sim \mathcal{N}(0, I)$
$y \leftarrow \text{U-Krig}(\tilde{y})$
for $i = 0, \ldots, N-1$ **do**
  $t_i = i/N$
  $x_{i+1} = x_i + \dfrac{1}{N}\, v_\theta(x_i, y, t_i)$
end for
return x_N

Figure 1: The Algorithm Architecture of Ki-OT-CFM.

Given a coarse-resolution observation $\tilde{y}$, we first apply Universal Kriging interpolation to construct a spatially coherent conditional prior y. Conditioned on this prior, a flow matching model learns a continuous vector field that transports Gaussian samples toward the target high-resolution SLA distribution.

To improve the geometry of probability paths, we further incorporate minibatch optimal transport (OT) coupling, which produces shorter and more consistent interpolation trajectories during training. In addition, a variogram regularization term is introduced to preserve the spatial dependence structures characteristic of oceanographic fields.

During inference, the learned vector field is integrated using a small number of Euler steps to efficiently generate high-resolution SLA realizations.

3.2 Kriging-Based Conditional Prior

Sea surface height fields exhibit strong spatial continuity and large-scale geophysical structures. Directly generating high-resolution samples from Gaussian noise alone is therefore highly under-constrained.

To provide physically meaningful spatial guidance, we employ Universal Kriging[1] to interpolate the coarse-resolution observation $\tilde{y}$ into a dense conditional map y. Unlike deterministic interpolation approaches such as bicubic upsampling, kriging explicitly models spatial covariance structures and produces geostatistically consistent estimates.

The interpolated field y is used as a condition throughout the generative process. At each time step, the evolving state x_t is concatenated with the kriging prior y and fed into the vector field network:

$$v_\theta(x_t, y, t).$$

This conditioning strategy preserves large-scale oceanographic structures while allowing the model to synthesize fine-scale variability.

3.3 Conditional Flow Matching

We build upon the Conditional Flow Matching (CFM) framework[1][2], which learns a continuous vector field transporting samples from a simple prior distribution to the target data distribution.

Let

$$z \sim \mathcal{N}(0, I)$$

denote a Gaussian latent variable and let x_0 denote the target high-resolution SLA field. A conditional probability path is defined using linear interpolation:

$$x_t = (1 - t)z + tx_0, \quad t \in [0, 1].$$

The corresponding target vector field is

$$u_t(x_t \mid x_0) = x_0 - z.$$

A U-Net parameterizes the learnable vector field $v_\theta(x_t, y, t)$, where the noisy state x_t and kriging condition map y are concatenated along the channel dimension as network input.

The standard flow matching objective is defined as

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,z,x_0}\left[\|v_\theta(x_t, y, t) - u_t\|_2^2\right].$$

3.4 Optimal-Transport Conditional Paths

Following Tong et al. [3], we employ minibatch optimal transport (OT) coupling to improve the geometry of conditional probability paths.

Instead of independently pairing Gaussian samples z with target samples x_0, OT computes a transport plan

$$\pi_{\mathrm{OT}}(z, x_0)$$

that minimizes the overall transport cost within each minibatch.

The resulting coupling produces shorter and more coherent interpolation trajectories:

$$x_t = (1-t)z + tx_0.$$

Compared with random pairing, OT coupling reduces trajectory crossing and simplifies the learned vector field, leading to more stable optimization and improved sample quality.

3.5 Variogram Regularization

Although flow matching learns the global data distribution, it does not explicitly enforce spatial dependence structures that are critical in geophysical fields.

To preserve spatial coherence, we adopted a variogram regularization term introduced in [1]. Given a reconstructed estimate $\hat{x}_0$, we compute its empirical variogram over a set of spatial lags and penalize discrepancies from the ground-truth variogram:

$$\mathcal{R}_V = \mathrm{MSE}\left(\gamma(\hat{x}_0) - \gamma(x_0)\right),$$

where $\gamma(\cdot)$ denotes the empirical variogram operator.

The final training objective is defined as

$$\mathcal{L} = \mathcal{L}_{\mathrm{FM}} + \lambda_V \mathcal{R}_V,$$

where λ_V controls the strength of the spatial regularization term.

3.6 Inference

At inference time, generation starts from Gaussian noise:

$$x_{t=0} \sim \mathcal{N}(0, I).$$

The learned vector field is then integrated from $t = 0$ to $t = 1$ conditioned on the kriging prior y. We employ Euler integration with $N = 50$ steps:

$$x_{i+1} = x_i + \frac{1}{N} v_\theta(x_i, y, t_i),$$

where

$$t_i = \frac{i}{N}.$$

Compared with diffusion-based approaches which require long reverse stochastic chains, the flow-based framework enables faster deterministic sampling.

4 Experiments

4.1 Experimental Setup

We evaluate the proposed Ki-OT-CFM framework on the task of sea level anomaly (SLA) downscaling over the Eastern North America (ENA) region (20°N–50°N, 80°W–40°W), as shown in Figure 2. The SLA data are obtained from the Copernicus Marine Service following the preprocessing protocol in prior work [1].

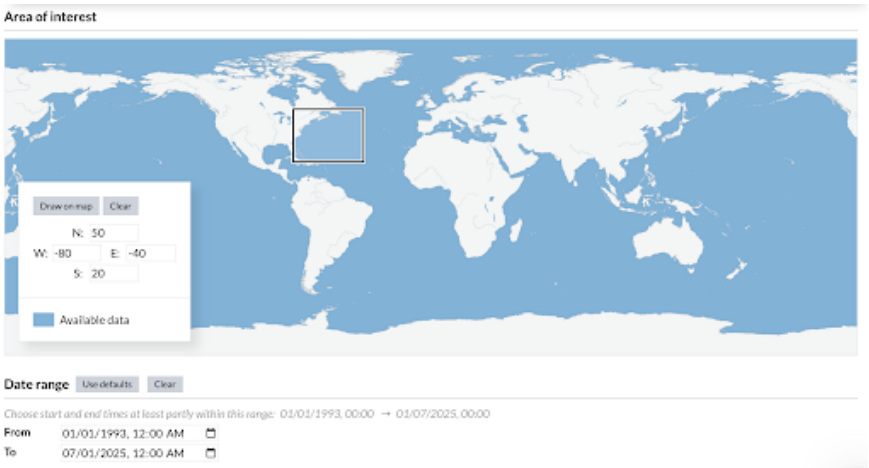


Figure 2: Area of interest (AOI) over the Eastern North America (ENA) region

The original SLA observations have a spatial resolution of $0.125° \times 0.125°$. Following previous studies, the data are downsampled to $0.25° \times 0.25°$ resolution as the high-resolution target, while an additional $1° \times 1°$ downsampling is used to construct coarse-resolution inputs, as shown in Figure 3. Universal Kriging (U-Krig) interpolation is then applied to the coarse observations to generate conditional maps used by the generative models, as shown in Figure 4.

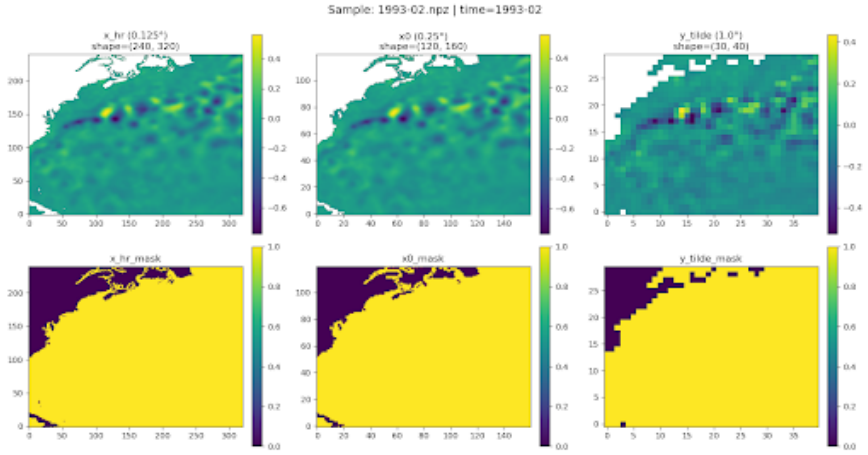


Figure 3: Downsampling to get coarse resolution map

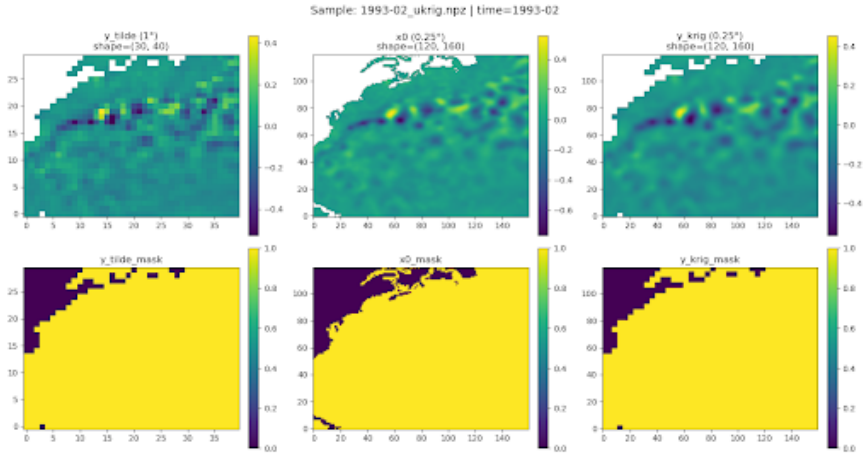


Figure 4: Universal Kriging to get interpolated conditional map

The training set contains monthly SLA fields from February 1993 to December 2013 (251 samples), while the test set contains monthly data from January 2014 to May 2023 (113 samples).

Both Ki-CDPM and Ki-OT-CFM were trained on A100 GPU in Google Colab. Ki-CDPM was trained for 500 epochs with a batch size of 16, a learning rate of 3×10^{-4}, and 1000 diffusion noise steps. Ki-OT-CFM was also trained for 500 epochs, using a batch size of 8 and a smaller learning rate of 1×10^{-4} for stable flow matching optimization. For both methods, the variogram regularization parameters were set to $\lambda_V = 0.1$ and $n_{\text{lags}} = 8$.

4.2 Baselines and Evaluation Metrics

We compare Ki-OT-CFM against both interpolation-based and diffusion-based baselines:

- Bicubic interpolation
- Universal Kriging (U-Krig)
- Ki-CDPM [1, 19, 20]
- Ki-OT-CFM

Performance is evaluated using the following metrics, which are fully illustrated in [1]:

- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)
- Pearson Correlation Coefficient (PCC)
- Continuous Ranked Probability Score (CRPS)

In addition, we compare computational efficiency in terms of training time and inference time.

4.3 Quantitative Results

Table 1 reports the quantitative evaluation results on the ENA test set.

Table 1: Quantitative evaluation results on the ENA test set

Method	**RMSE** $\downarrow$	**MAE** $\downarrow$	**PCC** $\uparrow$	**CRPS** $\downarrow$
Bicubic	0.0385	0.0246	0.9446	N/A
U-Krig	0.0443	0.0279	0.9323	N/A
Ki-CDPM	**0.0197**	**0.0136**	**0.9862**	**0.0100**
Ki-OT-CFM (Euler)	0.0260	0.0177	0.9757	0.0103
Ki-OT-CFM (dopri5)	0.0274	0.0188	0.9729	0.0104

The results show that Ki-OT-CFM improves over traditional interpolation approaches, achieving lower RMSE and MAE as well as higher PCC compared with Bicubic and U-Krig interpolation. These results indicate that the proposed conditional flow matching framework is able to recover finer spatial structures beyond deterministic interpolation methods.

Compared with Ki-CDPM, Ki-OT-CFM achieves slightly lower reconstruction quality in the current setting. However, the performance gap remains relatively small despite the substantially reduced sampling complexity of flow matching. Moreover, the current comparison is not fully controlled, since the two frameworks adopt different U-Net implementations and have not yet undergone extensive hyperparameter tuning.

Besides, Euler integration slightly outperforms the adaptive dopri5 solver in both accuracy and efficiency. This suggests that the dominant source of error lies in the learned vector field itself rather than numerical integration accuracy, implying that simple fixed-step solvers are sufficient for the current problem setting.

4.4 Computational Efficiency

Table 2 compares the computational cost of diffusion-based and flow-based approaches.

Table 2: Comparison of computational efficiency

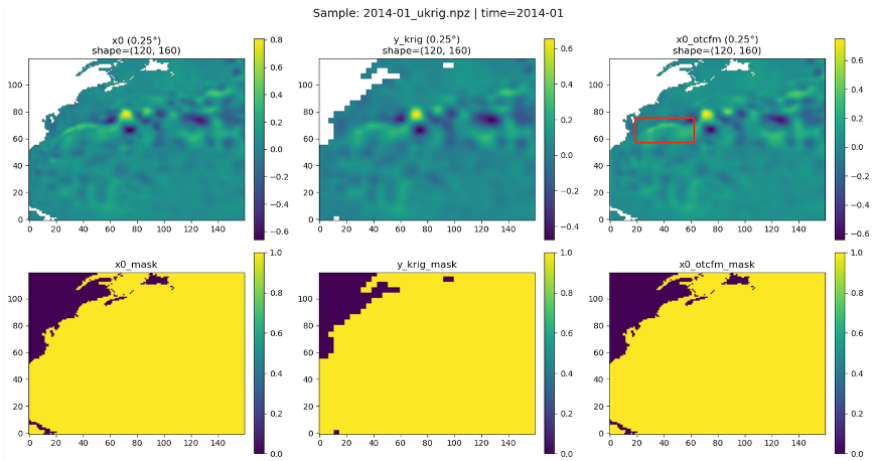
Method	**Training Time** (min / 500 epochs) ↓	**Inference Time** (s / img) ↓
Ki-CDPM	47.4	0.283
Ki-OT-CFM (Euler)	**15.1**	**0.194**
Ki-OT-CFM (dopri5)	**15.1**	0.448

Ki-OT-CFM demonstrates faster training and inference compared with Ki-CDPM. The acceleration mainly comes from the straight transport trajectories learned by optimal transport flow matching, which allow efficient deterministic sampling with a small number of Euler steps instead of long diffusion sampling chains.

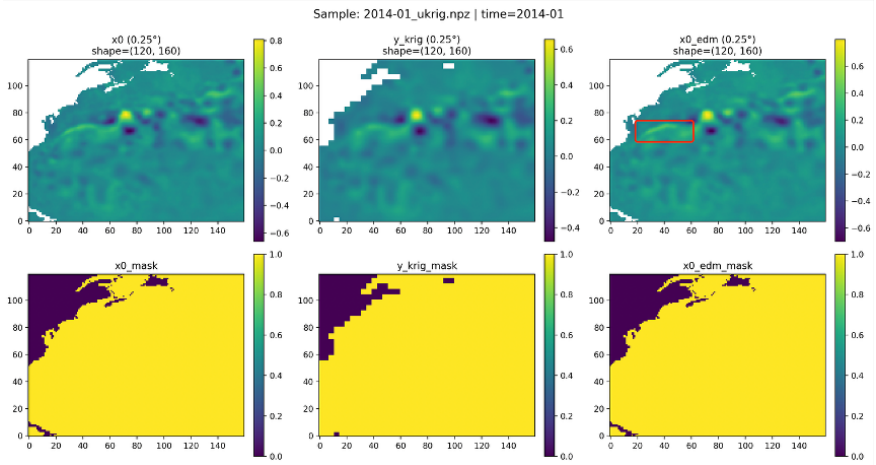
These results demonstrate the advantages of flow matching for climate downscaling: improved computational efficiency while maintaining competitive reconstruction quality.

4.5 Qualitative Results

Figure 5 presents qualitative comparisons between the generated SLA fields and the ground-truth observations. Both Ki-OT-CFM and Ki-CDPM are able to reconstruct the structures from coarse-resolution inputs and preserve the spatial distribution patterns. Compared with Ki-OT-CFM, Ki-CDPM produces sharper local details and clearer fine-scale variations in several regions, such as the area in the red box. This observation is also consistent with the quantitative performance reported in Table 1.



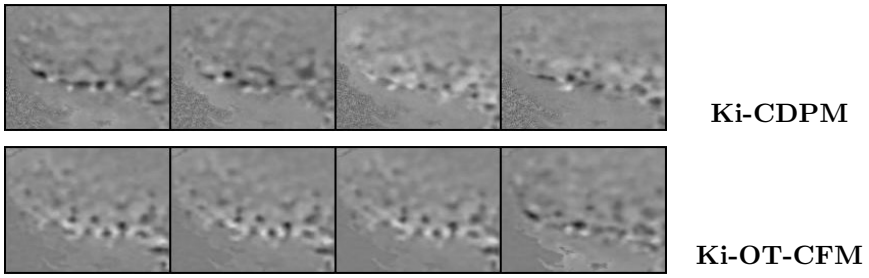
(a) Ki-OT-CFM downscaling results



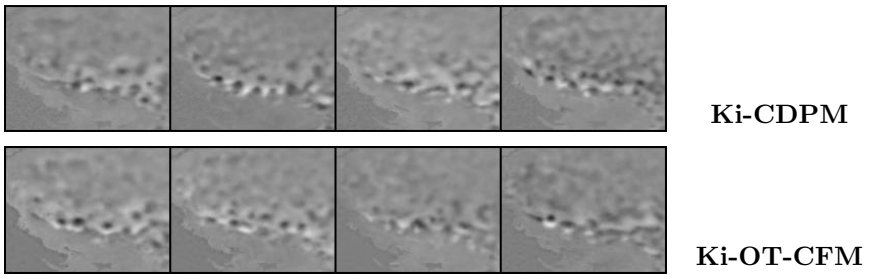
(b) Ki-CDPM downscaling results

Figure 5: Comparison of reconstructed SLA fields and corresponding ocean masks produced by Ki-OT-CFM and Ki-CDPM

We additionally compare intermediate sampling results during training, as shown in Figure 6. Ki-OT-CFM produces meaningful spatial structures at earlier training epochs, indicating faster convergence compared with Ki-CDPM. This observation is consistent with the theoretical motivation of OT-CFM, where optimal transport coupling reduces trajectory complexity and stabilizes vector field learning.



(a) Sampling results at 100 training epochs



(b) Sampling results at 490 training epochs

Figure 6: Comparison of intermediate sampling results between Ki-CDPM and Ki-OT-CFM at different training stages

5 Conclusion and Future Work

In this project, we propose Ki-OT-CFM, a kriging-informed optimal transport conditional flow matching framework for sea level anomaly downscaling. By combining Universal Kriging interpolation, optimal transport coupling, and variogram regularization within the conditional flow matching framework, the proposed method learns spatially coherent mappings from coarse-resolution observations to high-resolution fields.

Experimental results on the ENA dataset demonstrate that Ki-OT-CFM significantly improves over traditional interpolation-based approaches in both reconstruction accuracy and spatial detail recovery. Compared with diffusion-based methods, Ki-OT-CFM achieves substantially faster training and inference while

maintaining competitive quantitative performance. The results further suggest that optimal transport coupling helps stabilize vector field learning and enables efficient deterministic sampling using simple Euler integration.

Nevertheless, several limitations remain in the current study. First, the model architecture and hyperparameters have not yet been fully optimized, and the comparison with Ki-CDPM is not entirely controlled due to differences in implementation details such as the U-Net design. Second, the experiments are currently limited to synthetic downscaling settings on a single geographic region. In addition, the current framework mainly focuses on deterministic trajectory modeling and does not fully exploit stochastic transport dynamics.

Future work will focus on improving the model architecture and conducting more extensive hyperparameter tuning to further enhance generation quality. We also plan to investigate more flexible source distributions in OT-CFM, such as directly learning mappings from kriging-interpolated conditional fields to high-resolution observations. Moreover, stochastic extensions such as Schrödinger Bridge Conditional Flow Matching will be explored to better model uncertainty in climate downscaling. Finally, we aim to validate the proposed framework on larger-scale real-world climate datasets, including CMIP6 experiments and broader regional downscaling tasks.

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